Lesson 5.5 — Rational Exponents

Big idea: $x^{m/n}$ means the same thing as the nth root of x raised to the m. Rational exponents let you handle roots with the algebra rules you already know for exponents.

Quick review

Definition: $x^{1/n}$ = ■√x. $x^{m/n}$ = (■√x)^m = ■√(x^m).
Negative exponent: $x^{-p} = 1/x^p$.
Rules carry over: $x^a \cdot x^b = x^{a+b}$, $(x^a)^b = x^{ab}$, $x^a/x^b = x^{a-b}$.
Domain caution: for fractional powers with even denominators, x must be ≥ 0 to stay in the reals.

Worked example

Worked example. Simplify $16^{3/4} \cdot 16^{1/4}$.

$= 16^{3/4 + 1/4} = 16^1 = 16$.

Warm-ups

1. Rewrite $\sqrt{x}$ with a rational exponent.
2. Compute $8^{2/3}$.
3. Compute $25^{-1/2}$.

Practice

1. Simplify $27^{4/3}$.
2. Simplify $(x^{1/2})^4$.
3. Simplify $x^{2/3} \cdot x^{1/3}$.

Challenge

1. Simplify $(x^3y^2)^{1/2} / (x\,y)^{1/4}$, $x > 0$, $y > 0$.
2. Solve $x^{2/3} = 9$ for x (real solutions).

Answer key — Lesson 5.5

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $x^{1/2}$.
2. $(\blacksquare 8)^2 = 2^2 = 4$.
3. $1/\sqrt{}25 = 1/5$.

Practice

1. $(\blacksquare 27)^4 = 3^4 = 81$.
2. x^2.
3. $x^1 = x$.

Challenge

1. $x^{3/2}y^1 / (x^{1/4}y^{1/4}) = x^{3/2 - 1/4}y^{3/4} = x^{5/4}y^{3/4}$.
2. Raise both sides to 3/2: $x = 9^{3/2} = (\sqrt{}9)^3 = 27$. But also $x = (-27)$: $(-27)^{2/3} = (\blacksquare(-27))^2 = (-3)^2 = 9$. Both $x = 27$ and $x = -27$ work.